EC2L104 – Introduction to EDA Tools

LTspice Exercise Report

Name: Krishna Kaushik D Roll No.: 25EC01016

October 30, 2026

About this report

- **Problem:** my first results looked wrong (threshold near 0 V, perfectly flat saturation, current 6× too low).
- **Cause:** the card's model name `nmos` clashed with LTspice's built-in `NMOS`, so LTspice silently used its simple Level 1 model instead of the PTM BSIM4 card.
- **Fix:** renamed the models to `nmos0` / `pmos0`. All results below are from the BSIM4 card.

S1. Reading the Model Card

- $L_{\text{eff}} = L + XL - 2\,LINT, \quad W_{\text{eff}} = W - 2\,WINT, \quad C_{ox} = \varepsilon_0 \varepsilon_{rox}/TOXE$
- Card values: $XL = -30\,\text{nm}$, $LINT = 5.25\,\text{nm}$, $WINT = 5\,\text{nm}$, $TOXE = 1.85\,\text{nm}$

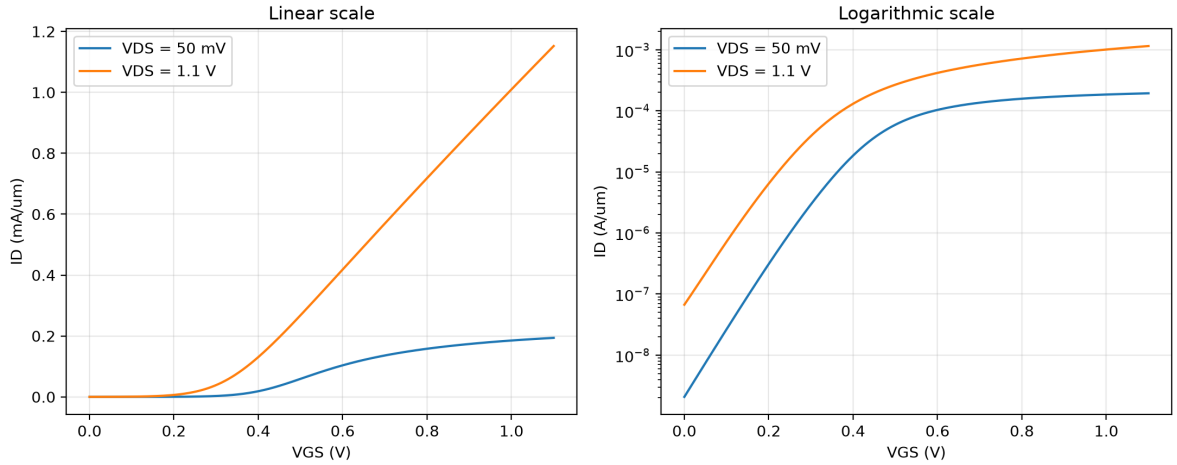
Quantity	$L = 65\,\text{nm}$	$L = 70\,\text{nm}$
L_{eff} (nm)	$65 - 30 - 10.5 = 24.5$	$70 - 30 - 10.5 = 29.5$
W_{eff} (μm), $W = 1\,\mu\text{m}$	0.99	0.99
C_{ox} ($\mu\text{F/cm}^2$)	1.87	1.87
I_D at $V_{GS} = V_{DS} = 1.1\,\text{V}$ (μA)	1152.29	1048.56

Observation

- I_D drops by 9.0 % when L goes from 65 to 70 nm.
- The drop is less than $1/L_{\text{eff}}$ predicts (ratio 0.910 against 0.831), because velocity saturation and RDSW limit the current.

S2. Transfer Characteristic (I_D–V_G)

- Simulated at $V_{DS} = 50\,\text{mV}$ and 1.1 V ($L = 65\,\text{nm}$), exported, plotted in Python.
- $V_{TH,\text{lin}}$: tangent at max g_m. $V_{TH,\text{cc}}$: V_{GS} at $I_D = 100\,\text{nA} \times W/L$. DIBL $= \Delta V_{TH,\text{cc}}/\Delta V_{DS}$.



I_D–V_{GS} at $V_{DS} = 50\,\text{mV}$ and $1.1\,\text{V}$, linear (left) and log (right).

Quantity	SPICE	Table 5.1	Difference
$V_{TH,\text{lin}}$ at 50 mV (V)	0.376	0.38	−1.1 %
$V_{TH,\text{cc}}$ at 50 mV (V)	0.270	0.27	0 %
$V_{TH,\text{cc}}$ at 1.1 V (V)	0.133	0.13	+2.3 %
DIBL (mV/V)	130.0	≈ 130	0 %
SS at 50 mV (mV/dec)	89.7	90	−0.3 %
SS at 1.1 V (mV/dec)	96.0	96	0 %
I_{ON} at 1.1 V (mA/μm)	1.152	1.15	+0.2 %
I_{OFF} at 1.1 V (nA/μm)	66.9	≈ 67	−0.1 %

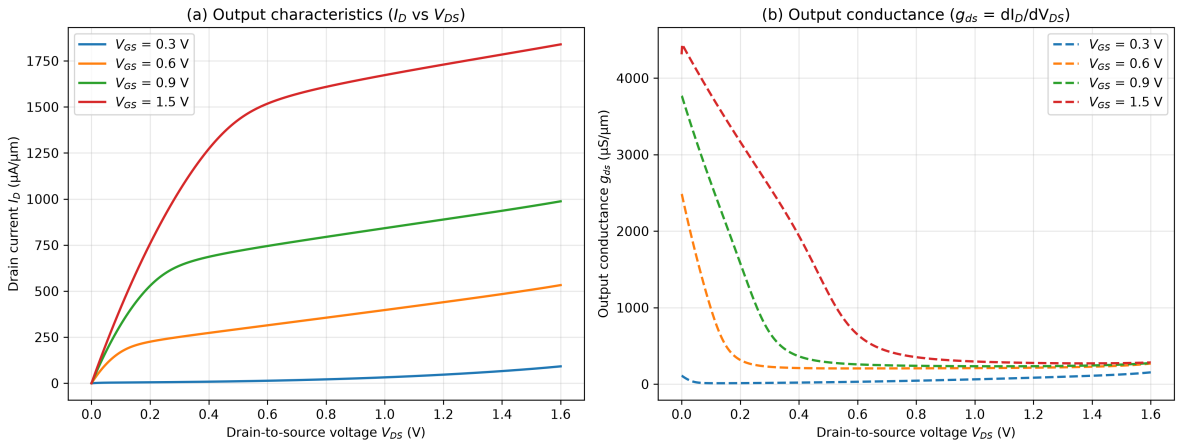
Observation

- $V_{TH,\text{cc}}$ falls from 0.270 to 0.133 V as V_{DS} rises: DIBL ≈ 130 mV/V (drain field lowers the source barrier in a short channel).
- $I_{OFF} = 67\,\text{nA}/\mu\text{m}$ at 1.1 V, about 30× the 50 mV value (about $2\,\text{nA}/\mu\text{m}$), from the same barrier lowering.
- SS = 90–96 mV/dec, above the ideal 60: depletion capacitance and charge sharing. Higher at 1.1 V because of DIBL.
- $V_{TH,\text{lin}}$ (0.376 V) > $V_{TH,\text{cc}}$ (0.270 V): max-g_m is in strong inversion, where mobility degradation and RDSW bend the curve.
- $I_{ON} = 1.152\,\text{mA}/\mu\text{m}$, set by velocity saturation. $I_{ON}/I_{OFF} = 1.7 \times 10^4$.

Conclusion: all values match Section 5.1 within about 2 %, so the card is used correctly.

S3. Output Characteristic (I_D–V_D)

- I_D–V_{DS} at $V_{GS} = 0.3, 0.6, 0.9, 1.5\,\text{V}$ (V_{DS} up to 1.6 V). g_{ds} from `np.gradient`, $r_o = 1/g_{ds}$ per μm of width.
- Saturation r_o at $V_{GS} = 1.5\,\text{V}$: about $\mathbf{3.6\,k\Omega \cdot \mu m}$ ($g_{ds} \approx 275\,\mu\text{S}/\mu\text{m}$ at $V_{DS} = 1.5\,\text{V}$).



Output characteristics (left) and g_{ds} (right).

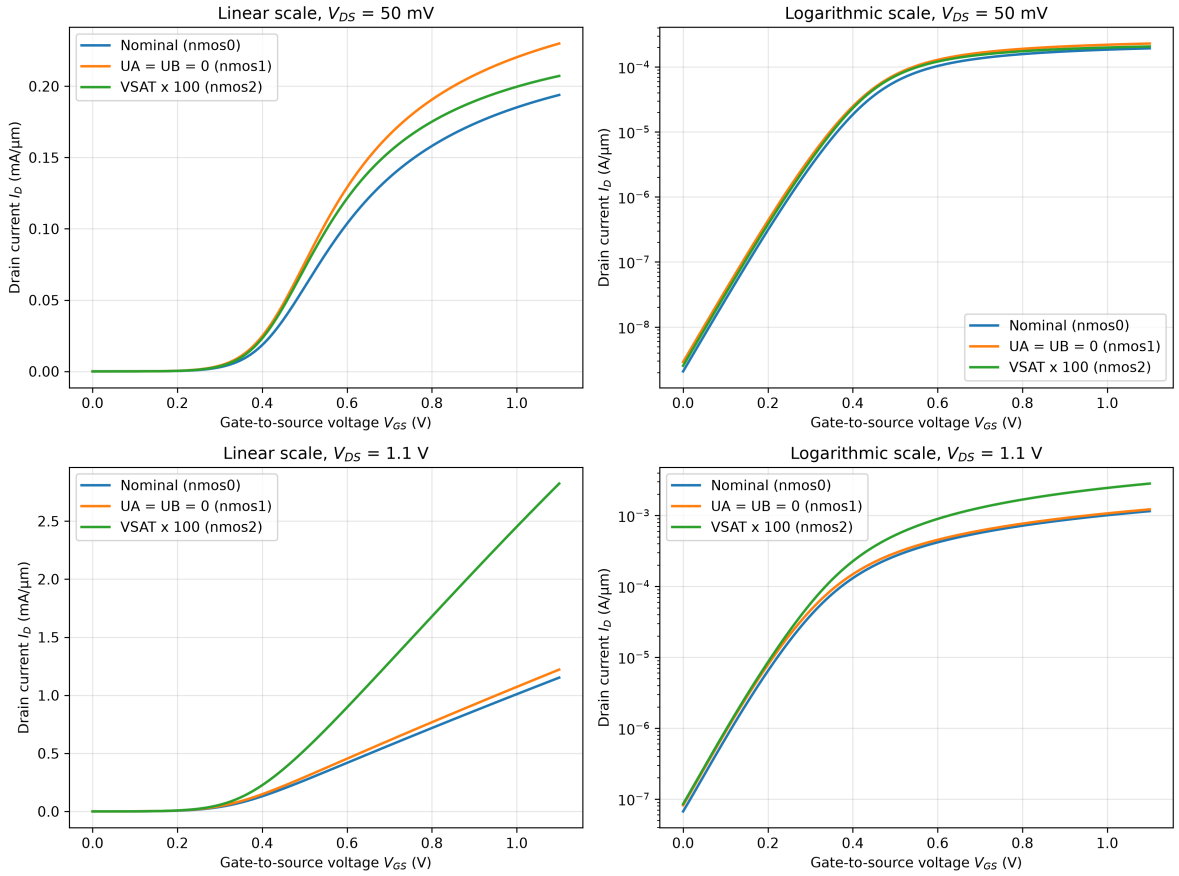
V_{GS} (V)	V_{DS} (V)	I_D (μA/μm)	g_{ds} (μS/μm)	r_o (k$\Omega \cdot \mu$m)
0.3	1.1	38.7	73.5	13.6
0.3	1.5	77.8	127.1	7.87
0.6	1.1	418.5	212.1	4.72
0.6	1.5	507.5	241.6	4.14
0.9	1.1	865.3	233.7	4.28
0.9	1.5	961.3	255.9	3.91
1.5	1.1	1701.9	285.0	3.51
1.5	1.5	1812.0	275.0	3.64

Observation

- Curves saturate early (knee near $V_{DS} \approx 0.6$–0.7 V at $V_{GS} = 1.5$ V, against $V_{GS} - V_{TH} \approx 1.2$ V): velocity saturation.
- g_{ds} does not reach zero (210–285 μS/μm for $V_{GS} \geq 0.6$ V), so r_o is 3.5–4.7 k$\Omega \cdot \mu$m: channel-length modulation (PCLM), DIBL and other drain-induced terms.
- r_o falls from 4.7 to 3.5 k$\Omega \cdot \mu$m as V_{GS} rises, since the saturation current grows faster than its slope.
- At $V_{GS} = 0.3$ V the device is near threshold and DIBL keeps raising the current, so those r_o values are not true saturation values.
- The 1.5 V rows are beyond the supply; they only show that g_{ds} has levelled off.

S4. Physics Check

- Reran S2 at $V_{DS} = 50$ mV and 1.1 V with three cards: nominal (`nmos0`), UA = UB = 0 (`nmos1`), VSAT × 100 (`nmos2`).
- UA = UB = 0 removes gate-field mobility degradation → **Enormal**.
- VSAT × 100 removes velocity saturation → **HighFieldSaturation**.



Nominal, UA = UB = 0 and VSAT × 100 at $V_{DS} = 50\,\text{mV}$ (left) and 1.1 V (right).

Case	V_{DS} (V)	$V_{TH,\text{cc}}$ (V)	I_{ON} (mA/µm)	I_{OFF} (nA/µm)
Nominal	0.05	0.270	0.194	2.07
UA = UB = 0	0.05	0.255	0.230 (+18.6 %)	2.89 (+39 %)
VSAT × 100	0.05	0.261	0.207 (+6.9 %)	2.53 (+22 %)
Nominal	1.1	0.133	1.152	66.9
UA = UB = 0	1.1	0.124	1.222 (+6.0 %)	81.6 (+22 %)
VSAT × 100	1.1	0.122	2.823 (+145 %)	85.2 (+27 %)

Observation

- **Enormal:** I_{ON} rises 18.6 % at 50 mV but only 6.0 % at 1.1 V. Mobility matters most at low V_{DS}.
- **HighFieldSaturation:** I_{ON} at 1.1 V goes from 1.152 to 2.823 mA/µm (×2.45), but only +6.9 % at 50 mV. The lateral field is high at 1.1 V ($V_{DS}/L_{\text{eff}} \approx 45\,\text{MV/m}$), so velocity saturation dominates there.
- I_{OFF} rises 22–39 % and $V_{TH,\text{cc}}$ drops 9–15 mV. This is not a real threshold change: the whole current scales up, so the constant-current crossing moves ($SS \times \log_{10} 1.22 \approx 8\,\text{mV}$, 9 mV observed).

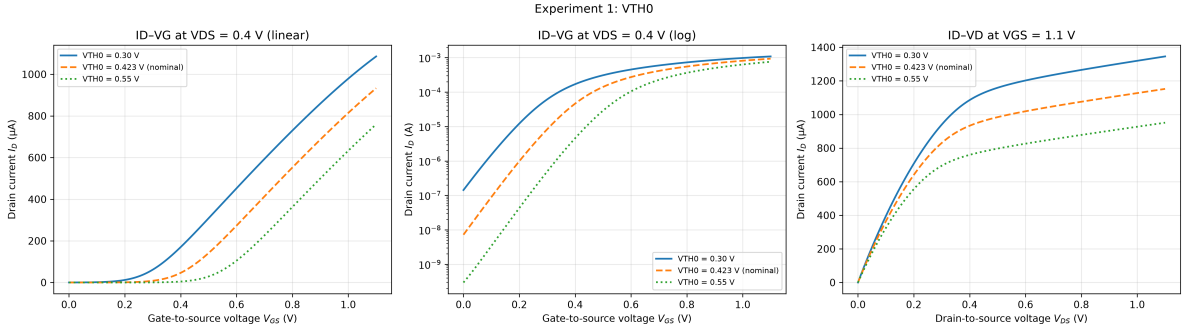
Conclusion: mobility degradation matters at low V_{DS}, velocity saturation at $V_{DS} = 1.1\,\text{V}$.

S5. Parametric Variation

- 14 parameters varied one at a time (low / nominal / high), others at nominal. Plots: I_D–V_{GS} at $V_{DS} = 0.4\,\text{V}$ (linear and log) and I_D–V_{DS} at $V_{GS} = 1.1\,\text{V}$.

- $V_{TH,\mathrm{cc}}$: $I_D = 100\,\mathrm{nA} \times W/L$ at $V_{DS} = 0.4\,\mathrm{V}$. I_{OFF}: $V_{GS} = 0$, $V_{DS} = 0.4\,\mathrm{V}$. I_{ON}: $V_{GS} = V_{DS} = 1.1\,\mathrm{V}$. All per drawn μm.
- Nominal: $V_{TH,\mathrm{cc}} = 0.220\,\mathrm{V}$, SS $= 90.7\,\mathrm{mV/dec}$, $I_{OFF} = 7.21\,\mathrm{nA}/\mu\mathrm{m}$, $I_{ON} = 1152\,\mu\mathrm{A}/\mu\mathrm{m}$.
- Groups: VTH0, TOXE, L, XL, LINT act through electrostatics; RDSW, VSAT, NGATE, U0 through transport; W, WINT, NSD, XJ have little effect.

Exp 1: VTH0

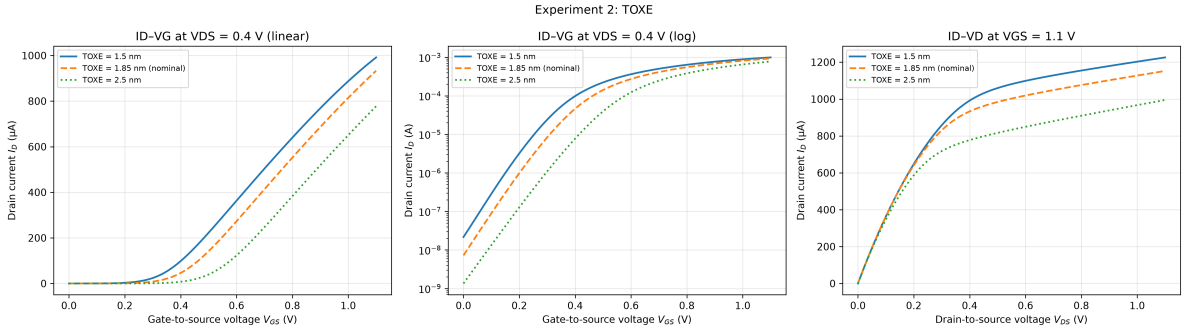


VTH0 = 0.30 / 0.423 / 0.55 V

VTH0	$V_{TH,\mathrm{cc}}$ (V)	SS (mV/dec)	I_{OFF} (nA/μm)	I_{ON} (μA/μm)
0.30 V	0.101	93.8	142.9	1345.6
0.423 V (nom.)	0.220	90.7	7.21	1152.3
0.55 V	0.349	90.3	0.295	951.1

- $V_{TH,\mathrm{cc}}$ follows VTH0 almost one for one (-119 and $+129\,\mathrm{mV}$).
- I_{OFF} changes about 20× each way (one decade per 90 mV); I_{ON} only $\pm 17\,\%$ because 0.12 V is small against the overdrive.
- SS unchanged: VTH0 is a pure offset.

Exp 2: TOXE

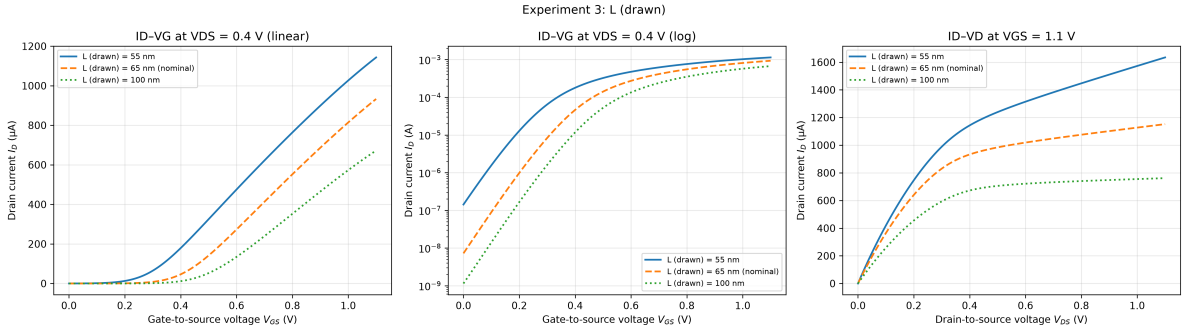


TOXE = 1.5 / 1.85 / 2.5 nm

TOXE	$V_{TH,\mathrm{cc}}$ (V)	SS (mV/dec)	I_{OFF} (nA/μm)	I_{ON} (μA/μm)
1.5 nm	0.168	88.1	21.6	1225.7
1.85 nm (nom.)	0.220	90.7	7.21	1152.3
2.5 nm	0.316	99.9	1.35	995.1

- C_{ox} falls from 2.30 to 1.38 μF/cm^2 (1.5 to 2.5 nm), so gate control weakens: SS up, I_{ON} −14 %, $V_{TH,\mathrm{cc}}$ +96 mV, I_{OFF} down 5×.
- Part of the V_{TH} shift is a model effect: BSIM4 scales K1 and K2 by TOXE/TOXM (TOXM stays 1.85 nm) while VTH0 is fixed.

Exp 3: L (drawn)

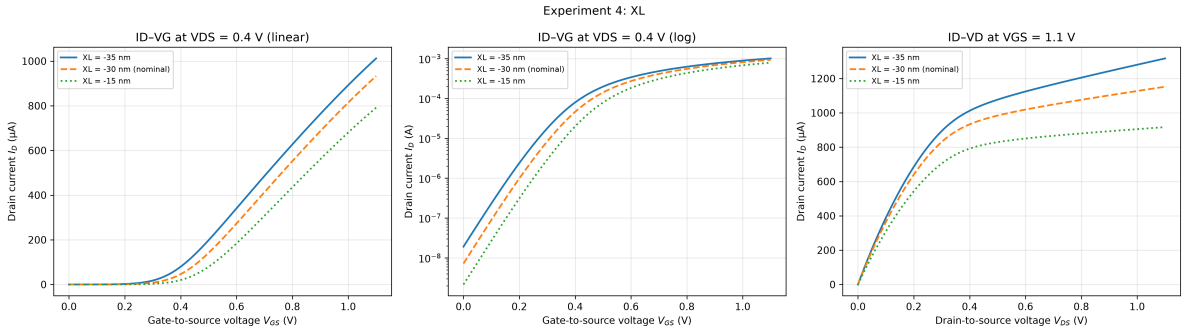


$L = 55\ /\ 65\ /\ 100$ nm

L	$V_{TH,\mathrm{cc}}$ (V)	SS (mV/dec)	I_{OFF} (nA/μm)	I_{ON} (μA/μm)
55 nm	0.106	92.8	142.5	1634.7
65 nm (nom.)	0.220	90.7	7.21	1152.3
100 nm	0.278	91.0	1.14	761.8

- L_{eff} = 14.5, 24.5, 59.5 nm. Shorter channel: threshold roll-off (charge sharing, DIBL), so $V_{TH,\mathrm{cc}}$ −114 mV and I_{OFF} up 20×.
- I_{ON} +42 % / −34 %, weaker than $1/L_{\mathrm{eff}}$ because of velocity saturation and RDSW.
- Roll-off flattens for long channels (only +58 mV from 65 to 100 nm).

Exp 4: XL

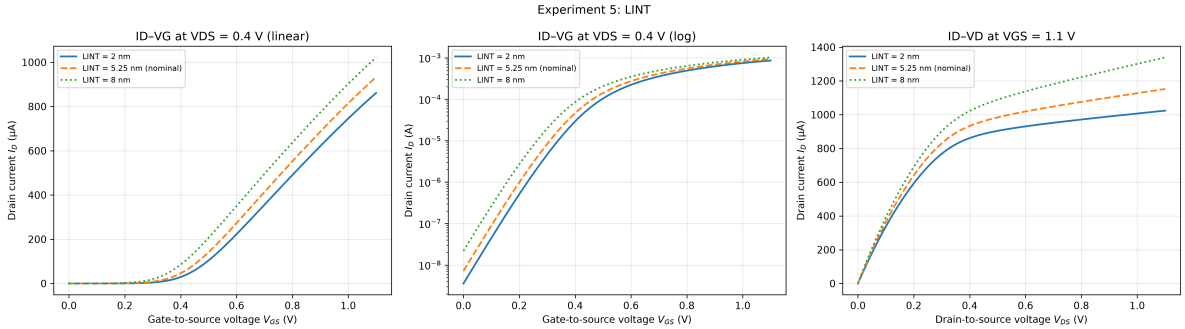


XL $= -35\ /\ -30\ /\ -15$ nm

XL	$V_{TH,\mathrm{cc}}$ (V)	SS (mV/dec)	I_{OFF} (nA/μm)	I_{ON} (μA/μm)
−35 nm	0.180	90.9	19.1	1317.0
−30 nm (nom.)	0.220	90.7	7.21	1152.3
−15 nm	0.270	90.8	2.15	916.8

- XL only changes L_{eff} (19.5, 24.5, 39.5 nm), so it behaves like Exp 3: shorter L_{eff} gives lower $V_{TH,\text{cc}}$, higher I_{OFF} and I_{ON}.

Exp 5: LINT

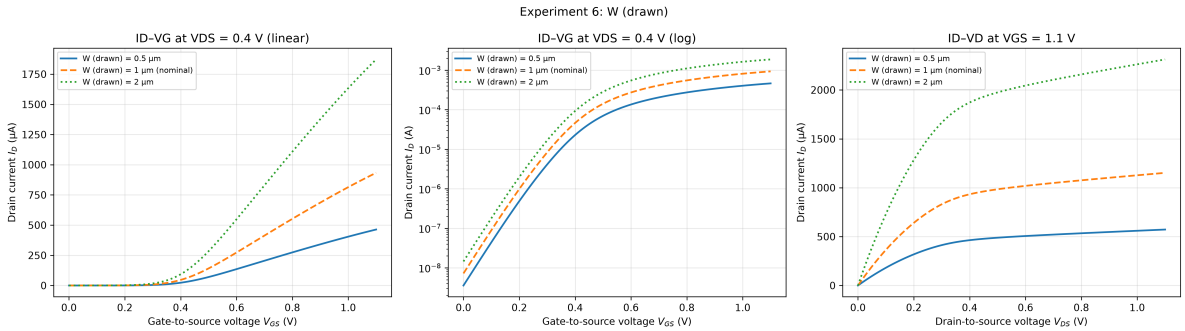


LINT = 2 / 5.25 / 8 nm

LINT	$V_{TH,\text{cc}}$ (V)	SS (mV/dec)	I_{OFF} (nA/μm)	I_{ON} (μA/μm)
2 nm	0.248	90.4	3.58	1024.0
5.25 nm (nom.)	0.220	90.7	7.21	1152.3
8 nm	0.175	91.0	22.0	1339.3

- Larger LINT means more S/D diffusion under the gate, so L_{eff} is shorter (31, 24.5, 19 nm).
- $V_{TH,\text{cc}}$ −45 mV, I_{OFF} up 3×, I_{ON} +16 % at 8 nm. LINT = 8 nm gives nearly the same result as XL = −35 nm.

Exp 6: W (drawn)

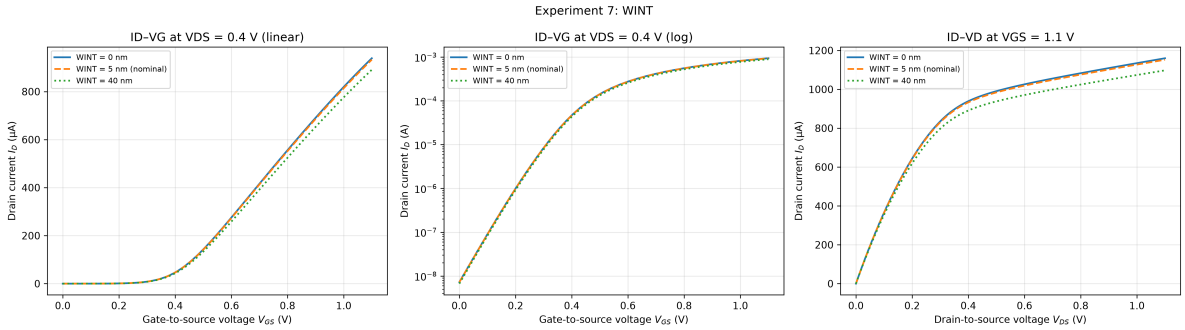


W = 0.5 / 1 / 2 μm

W	$V_{TH,\text{cc}}$ (V)	SS (mV/dec)	I_{OFF} (nA/μm)	I_{ON} (μA/μm)
0.5 μm	0.220	90.7	7.13	1144.6
1 μm (nom.)	0.220	90.7	7.21	1152.3
2 μm	0.219	90.7	7.24	1156.1

- Absolute current scales with W_{eff} (0.49, 0.99, 1.99 μm), about ×0.5, ×1, ×2. Per μm it is almost constant.
- V_{TH} and SS do not depend on width here (DVT0W, DVT1W, K3 = 0). The tiny I_{ON} change comes from the fixed 10 nm offset ($2\,WINT$).

Exp 7: WINT

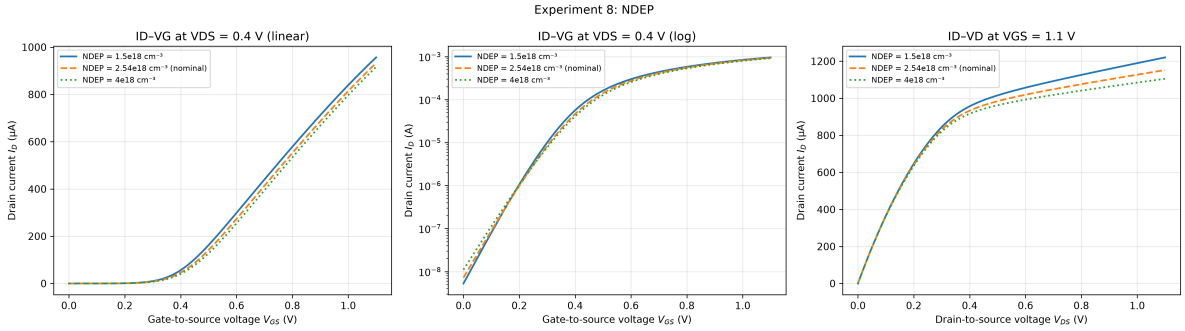


WINT = 0 / 5 / 40 nm

WINT	$V_{TH,\mathrm{cc}}$ (V)	SS (mV/dec)	I_{OFF} (nA/μm)	I_{ON} (μA/μm)
0 nm	0.219	90.7	7.28	1159.9
5 nm (nom.)	0.220	90.7	7.21	1152.3
40 nm	0.223	90.7	6.69	1097.6

- WINT only narrows the device: $W_{\mathrm{eff}} = 1.00, 0.99, 0.92\,\mu$m. At 40 nm, I_{ON} $-4.7\,\%$ and I_{OFF} $-7\,\%$ per drawn μm.
- $V_{TH,\mathrm{cc}}$ moves only 3 mV, so it is essentially a loss of width.

Exp 8: NDEP

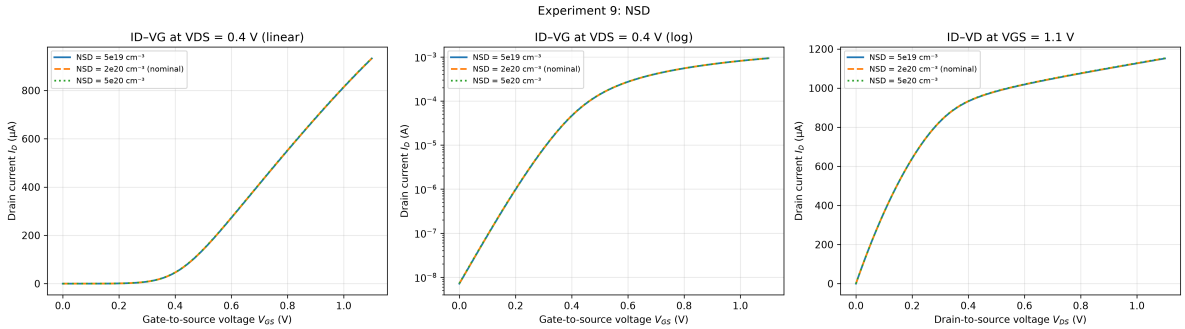


NDEP = 1.5×10^{18} / 2.54×10^{18} / $4 \times 10^{18}\,\mathrm{cm}^{-3}$

NDEP	$V_{TH,\mathrm{cc}}$ (V)	SS (mV/dec)	I_{OFF} (nA/μm)	I_{ON} (μA/μm)
1.5×10^{18}	0.215	84.0	5.28	1220.5
2.54×10^{18} (nom.)	0.220	90.7	7.21	1152.3
4×10^{18}	0.218	98.1	11.06	1105.5

- Matches physics: more doping raises SS (84 $\rightarrow$ 98 mV/dec) and lowers I_{ON} ($+5.9\,\%$, $-4.1\,\%$).
- Does *not* match: $V_{TH,\mathrm{cc}}$ does not rise and I_{OFF} increases. VTH0 is a fixed input to the card, so NDEP does not set the threshold in SPICE. It only acts through the depletion width (SS) and short-channel terms.
- With V_{TH} fixed, a larger SS means more current at $V_{GS} = 0$, hence the higher I_{OFF}.

Exp 9: NSD

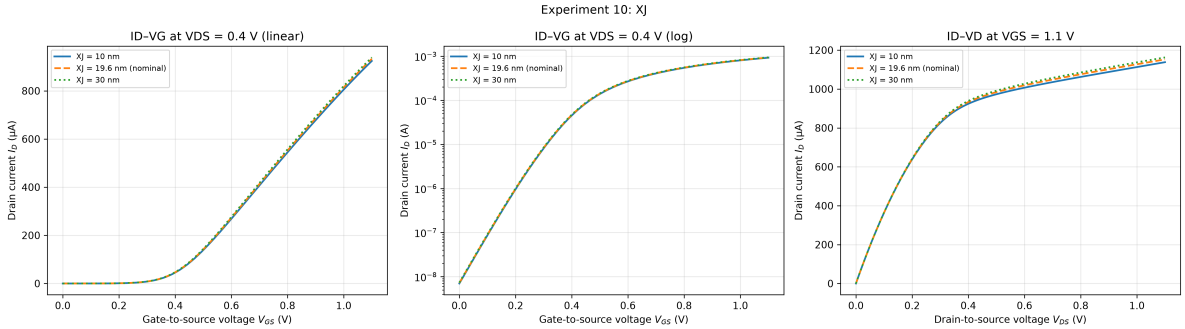


$$\text{NSD} = 5 \times 10^{19} \ / \ 2 \times 10^{20} \ / \ 5 \times 10^{20}\,\text{cm}^{-3}$$

NSD	$V_{TH,\text{cc}}$ (V)	SS (mV/dec)	I_{OFF} (nA/μm)	I_{ON} (μA/μm)
5×10^{19}	0.220	90.8	7.14	1151.6
2×10^{20} (nom.)	0.220	90.7	7.21	1152.3
5×10^{20}	0.219	90.7	7.25	1152.7

- Almost no effect ($< 1\,\%$). In a real device NSD sets series resistance, but here RDSW does that.
- NSD only enters through a logarithmic built-in-potential term, so a tenfold change moves V_{TH} by about 1 mV.

Exp 10: XJ

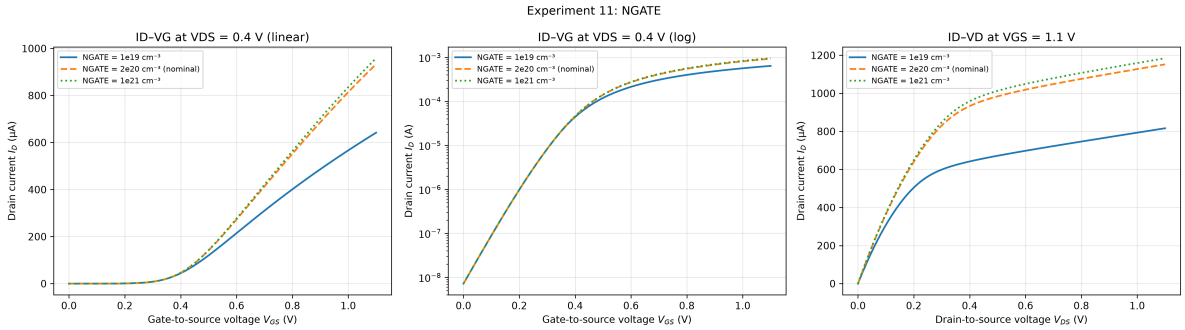


$$\text{XJ} = 10 \ / \ 19.6 \ / \ 30\,\text{nm}$$

XJ	$V_{TH,\text{cc}}$ (V)	SS (mV/dec)	I_{OFF} (nA/μm)	I_{ON} (μA/μm)
10 nm	0.221	90.8	7.00	1138.1
19.6 nm (nom.)	0.220	90.7	7.21	1152.3
30 nm	0.219	90.7	7.35	1162.5

- Direction is as expected (deeper junction, worse short-channel effects), but the size is tiny (I_{ON} $\pm 1\,\%$, I_{OFF} $\pm 3\,\%$).
- In the card XJ only enters the threshold correction, not the S/D resistance, so SPICE is much less sensitive than TCAD would be.

Exp 11: NGATE

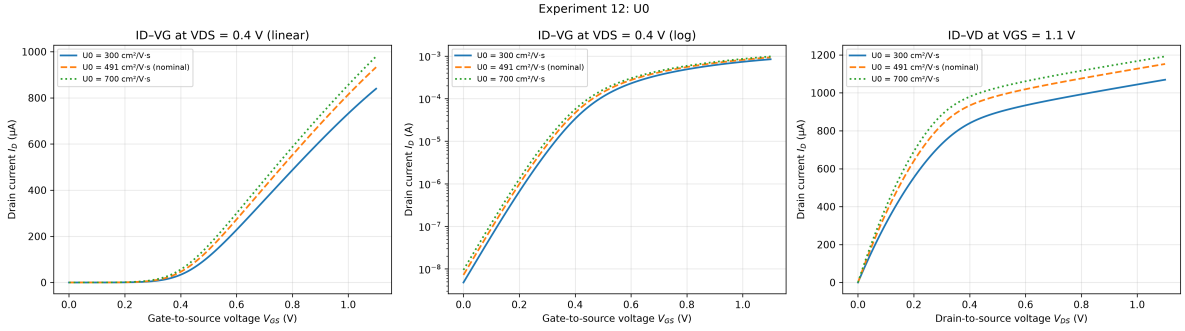


$$\text{NGATE} = 1\times10^{19} \;/\; 2\times10^{20} \;/\; 1\times10^{21}\,\text{cm}^{-3}$$

NGATE	$V_{TH,\text{cc}}$ (V)	SS (mV/dec)	I_{OFF} (nA/μm)	I_{ON} (μA/μm)
1×10^{19}	0.220	90.5	7.19	816.5
2×10^{20} (nom.)	0.220	90.7	7.21	1152.3
1×10^{21}	0.220	90.9	7.22	1184.3

- Poly-gate depletion: a lightly doped gate depletes at high field and adds to the effective oxide thickness, so I_{ON} drops 29 % at 10^{19}.
- $V_{TH,\text{cc}}$, SS and I_{OFF} do not change because the poly is not depleted at low V_{GS}.
- Nominal is already slightly depleted, so 10^{21} gives only +2.8 %. Agrees with device physics.

Exp 12: U0

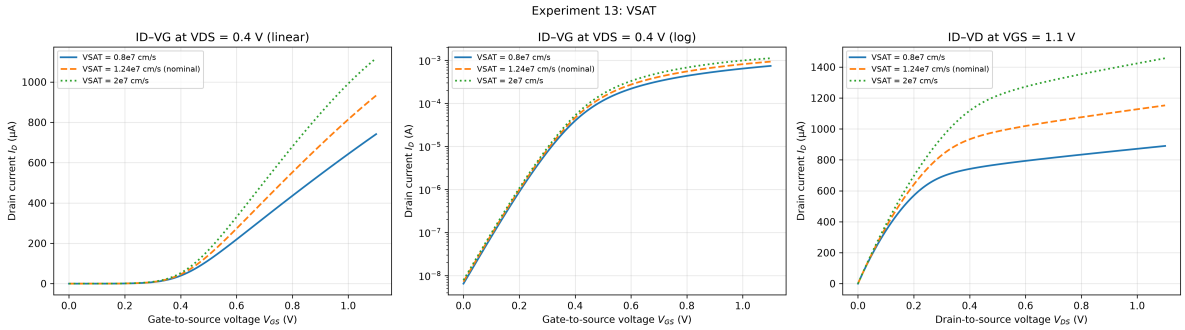


$$\text{U0} = 300 \;/\; 491 \;/\; 700\,\text{cm}^2/\text{V}\cdot\text{s}$$

U0	$V_{TH,\text{cc}}$ (V)	SS (mV/dec)	I_{OFF} (nA/μm)	I_{ON} (μA/μm)
300	0.238	90.9	4.79	1069.3
491 (nom.)	0.220	90.7	7.21	1152.3
700	0.208	90.6	9.46	1192.1

- $V_{TH,\text{cc}}$ moves (+18, −12 mV) only because the constant-current criterion samples a current that scales with mobility (expected +19, −14 mV). It is not a real threshold change.
- I_{OFF} scales roughly with mobility. I_{ON} changes only −7.2 % / +3.5 % for −39 % / +43 % mobility, because velocity saturation and RDSW limit it (same as S4).

Exp 13: VSAT

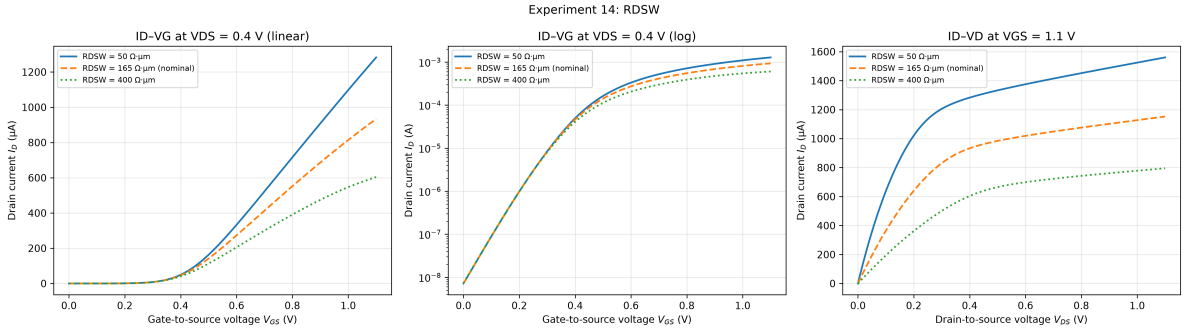


$$\text{VSAT} = 0.8 \times 10^7 \;/\; 1.24 \times 10^7 \;/\; 2 \times 10^7 \,\text{cm/s}$$

VSAT	$V_{TH,\text{cc}}$ (V)	SS (mV/dec)	I_{OFF} (nA/μm)	I_{ON} (μA/μm)
0.8×10^7	0.225	90.8	6.47	890.2
1.24×10^7 (nom.)	0.220	90.7	7.21	1152.3
2×10^7	0.216	90.7	7.82	1457.0

- I_{ON} changes $-22.7\,\%$ / $+26.4\,\%$: velocity saturation is the main limit on drive current here (as in S4).
- $V_{TH,\text{cc}}$ and I_{OFF} shift only slightly ($+5$, $-4\,\text{mV}$), the same constant-current effect as Exp 12 but weaker at low V_{DS}.

Exp 14: RDSW



$$\text{RDSW} = 50 \;/\; 165 \;/\; 400\,\Omega \cdot \mu\text{m}$$

RDSW	$V_{TH,\text{cc}}$ (V)	SS (mV/dec)	I_{OFF} (nA/μm)	I_{ON} (μA/μm)
50	0.220	90.7	7.17	1560.3
165 (nom.)	0.220	90.7	7.21	1152.3
400	0.220	90.7	7.27	795.2

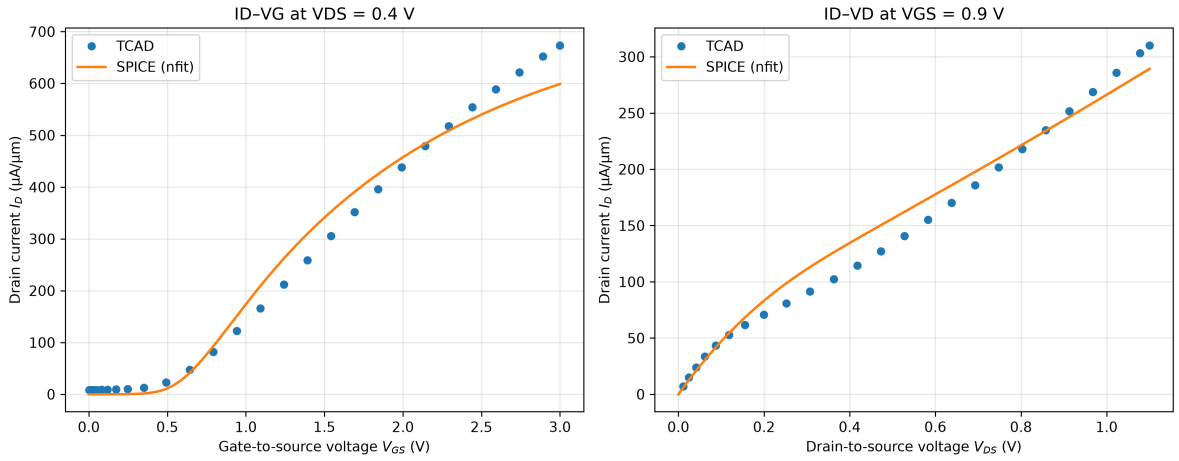
- Only I_{ON} changes ($+35\,\%$ / $-31\,\%$): the voltage lost across the series resistance is $I \times R$, which matters only at high current.
- At nominal, $1.15\,\text{mA}$ through about $167\,\Omega$ drops about $0.19\,\text{V}$ of the $1.1\,\text{V}$ supply.

S6. Approximate Match to TCAD

- Copied the PTM card to a new model `nfit` ($L = 70\,\text{nm}$, $W = 1\,\mu\text{m}$). Module 6 Step 0 first: $TOXE = TOXP = TOXM = 5\,\text{nm}$, $DTOX = 0$, $IGCMOD = IGBMOD = 0$.
- Starting point (PTM values): mean error 111 % (I_D–V_{GS}) and 139 % (I_D–V_{DS}); $V_{TH} = 0.410\,\text{V}$ against TCAD 0.561 V.

What I changed in each round (mean error: I_D–V_{GS} / I_D–V_{DS})

1. VTH0 0.423 → 0.57, U0 0.0491 → 0.027. Error 55 % / 40 %, $V_{TH} = 0.590\,\text{V}$.
2. VTH0 → 0.54, RDSW 165 → 400. Error 22 % / 34 %, $V_{TH} = 0.537\,\text{V}$. I_D–V_{GS} improved a lot, I_D–V_{DS} barely.
3. VTH0 → 0.56, U0 → 0.018, RDSW → 200. Error 27 % / 24.5 %, $V_{TH} = 0.597\,\text{V}$.
4. VTH0 → 0.54, U0 → 0.012, VSAT 124340 → 250000, RDSW → 280. Error 9.7 % / 17.1 %, $V_{TH} = 0.587\,\text{V}$. Best I_D–V_{GS}, but I_D–V_{DS} still low at high V_{DS}.
5. ETA0 0.0058 → 0.02, PCLM 0.04 → 0.3. Too strong: error rose to 30 % / 75 % and V_{TH} fell to 0.356 V.
6. Backed off: ETA0 → 0.010, PCLM → 0.12, VTH0 → 0.58. Error **11.8 % / 9.7 %**, $V_{TH} = 0.555\,\text{V}$. Kept as final.



TCAD (markers) and SPICE `nfit` (lines): I_D–V_{GS} at $V_{DS} = 0.4\,\text{V}$ (left), I_D–V_{DS} at $V_{GS} = 0.9\,\text{V}$ (right).

Parameter	PTM value	My value
VTH0 (V)	0.423	0.58
U0 (m^2/V·s)	0.0491	0.012
VSAT (m/s)	124340	250000
ETA0	0.0058	0.010
PCLM	0.04	0.12
RDSW ($\Omega \cdot \mu\text{m}$)	165	280

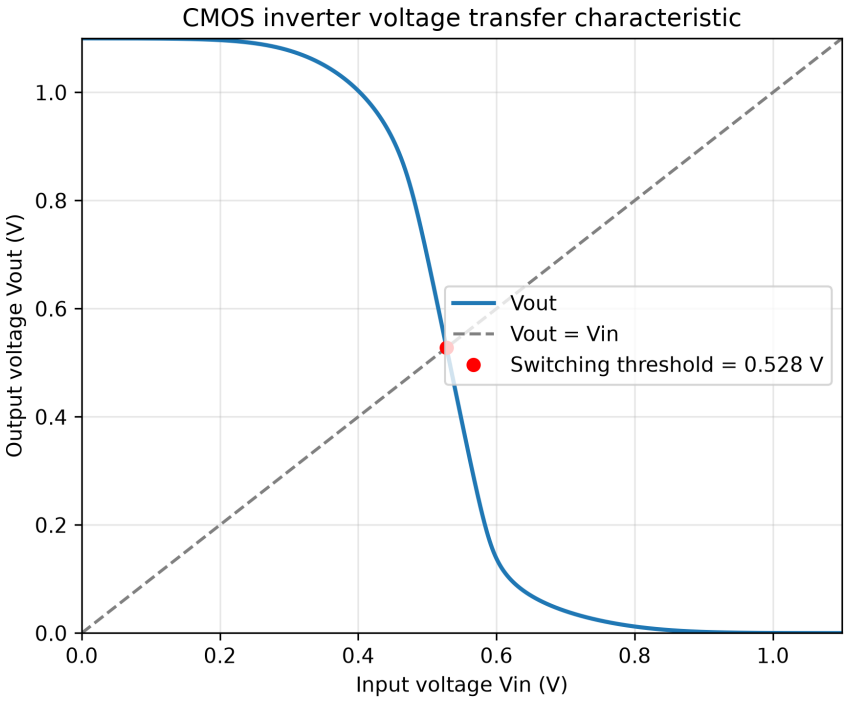
Curve	Mean error (%)	Max error (%)
I_D–V_{GS} at $V_{DS} = 0.4\,\text{V}$	11.8	22.9
I_D–V_{DS} at $V_{GS} = 0.9\,\text{V}$	9.7	21.9

Observation

- $V_{TH,\mathrm{lin}}$ (max g_m): SPICE 0.555 V, TCAD 0.561 V.
- Higher VTH0 sets the TCAD threshold. Lower U0 and higher RDSW bring the current down and make I_D–V_{GS} nearly linear at high V_{GS}. Higher VSAT, PCLM and ETA0 keep I_D–V_{DS} rising instead of flattening.
- RDSW = 240 gave almost the same fit (max errors 25.6 % / 23.8 %), so I kept 280.
- Remaining error is at the ends of the sweep (SPICE about 16 % below TCAD at $V_{GS} = 3$ V); one parameter set cannot match both curves perfectly.

S7. CMOS Inverter Simulation

- PTM cards `nmos0` / `pmos0`, $V_{DD} = 1.1$ V, $L = 65$ nm. nMOS $W = 1\,\mu$m, pMOS $W = 2\,\mu$m. Gates tied as V_{in}, drains as V_{out}, V_{in} swept 0 to 1.1 V.



V_{out} against V_{in} for the CMOS inverter.

Why is it called an inverter?

Input LOW ($V_{in} = 0$)

- nMOS is OFF, pMOS is ON.
- Output is HIGH: $V_{out} = 1.1\,\mathrm{V} = V_{DD}$.

Input HIGH ($V_{in} = 1.1$ V)

- nMOS is ON, pMOS is OFF.
- Output is LOW: $V_{out} \approx 0$ V.

Switching point

- Near the switching point both transistors conduct (both in saturation).

- The output changes quickly: the transition happens between about 0.4 and 0.65 V, so it is sharp.
- HIGH input gives LOW output, LOW input gives HIGH output: the circuit **inverts** the input.

Power consumption

- In steady state only one transistor is ON, so almost no current flows from supply to ground.
- Static power is therefore very low.
- During switching both conduct for a short time, giving a brief rise in current and power.

Switching voltage

- V_M (where $V_{out} = V_{in}$) = $\mathbf{0.528\,V}$, slightly below $V_{DD}/2 = 0.55\,\mathrm{V}$.
- The pMOS is weaker than the nMOS because holes have lower mobility. Even with $W = 2\,\mu\mathrm{m}$ the pMOS does not fully match the nMOS.